Quantum Mechanics II — Exercise Sheet 9

Weyl operators and phase space

Variant: Questions only

Exercise 1 (A closed displacement path). Use the convention $W(q,p) = e^{-\mathrm{i}qp/(2\hbar)}V(p)U(q)$. (i) Compute $W(q,0)W(0,p)W(-q,0)W(0,-p)$. Check the result by acting on a Schwartz function. (ii) Explain which phase is observable if this path is compared coherently with doing nothing.

Exercise 2 (A reducible Weyl pair). On $L^2(\mathbb{R}) \otimes \mathbb{C}^2$ let $U(a) = U_{\mathrm{line}}(a) \otimes \mathbf{1}$ and $V(b) = V_{\mathrm{line}}(b) \otimes \mathbf{1}$. Check Weyl's relation and strong continuity. Give a nontrivial closed subspace invariant under both groups and a nonscalar bounded operator commuting with them. Explain why the irreducible Stone–von Neumann theorem does not apply.

Exercise 3 (The Wigner function of a Gaussian). For $s > 0$ let $\psi(x) = (\pi s^2)^{-1/4} e^{-x^2/(2s^2)}$. Define its Wigner function by

$$F_\psi(q,p) = \frac{1}{2\pi\hbar} \int_{\mathbb{R}} e^{-\mathrm{i}py/\hbar} \psi(q + y/2)\overline{\psi(q - y/2)}\, \mathrm{d}y.$$

Compute F_ψ and its two marginals. Use the Gaussian integral

$$\int e^{-ay^2 - \mathrm{i}by}\, \mathrm{d}y = \sqrt{\pi/a}\, e^{-b^2/(4a)}$$

for $a > 0$. Does positivity for this one state establish a joint position and momentum observable?

Exercise 4 (A displacement through one oscillator period). Let H be the oscillator Hamiltonian, $U_t = e^{-\mathrm{i}tH/\hbar}$, and let $W(q,p)$ act on the oscillator ground state. Using its coherent-state interpretation, find $\langle Q \rangle_t$ and $\langle P \rangle_t$. Determine $U_{2\pi/\nu} W(q,p) h_0$ as a vector, not only as a physical state.